Abstract

Railway safety is an important concern, as problems such as accidents on platforms, damage to railway tracks, and the gradual deterioration of bridges can put both passengers and infrastructure at risk. This project proposes a smart monitoring system that helps detect potential safety issues across different parts of a railway network before they develop into serious problems.The system uses multiple monitoring units placed at important locations, such as railway platforms, tracks, and bridges. These units collect information about local conditions and process the data nearby, allowing the system to respond quickly to potential risks. Piezoelectric energy harvesting is also incorporated to generate small amounts of electrical energy from vibrations, helping power the monitoring units and reducing their dependence on external power sources. For bridges, continuous structural health monitoring helps track changes in load and stress that may indicate possible damage or weakness. All the collected information is brought together through a centralized digital twin interface, which provides a clear overview of the railway infrastructure. Instead of transmitting large amounts of raw data, the system sends only the essential information needed for monitoring and decision-making. This helps reduce communication delays and makes the system more efficient. Overall, the proposed system aims to provide a practical, affordable, and energy-efficient approach to railway safety. By combining real-time monitoring, local data processing, and energy harvesting, it offers a way to improve infrastructure management, identify potential risks early, and reduce the need for frequent manual inspections. with this abstract write problem, solution and innovative details.